

INVENTORY CONTROL SYSTEM MATERIAL REQUIREMENT PROCESSING

PROGRAM

```
SET SERVEROUTPUT ON;  
SET LINESIZE 120;  
SET PAGESIZE 50;
```

1. DROP OLD TABLES

```
BEGIN  
EXECUTE IMMEDIATE 'DROP TABLE Purchase_Order_Details CASCADE CONSTRAINTS';  
EXCEPTION  
WHEN OTHERS THEN NULL;  
END;  
/
```

```
BEGIN  
EXECUTE IMMEDIATE 'DROP TABLE Purchase_Order CASCADE CONSTRAINTS';  
EXCEPTION  
WHEN OTHERS THEN NULL;  
END;  
/
```

```
BEGIN  
EXECUTE IMMEDIATE 'DROP TABLE Requirement_Details CASCADE CONSTRAINTS';  
EXCEPTION  
WHEN OTHERS THEN NULL;  
END;  
/
```

```
BEGIN  
EXECUTE IMMEDIATE 'DROP TABLE Material_Requirement CASCADE CONSTRAINTS';  
EXCEPTION  
WHEN OTHERS THEN NULL;  
END;  
/
```

```
BEGIN  
EXECUTE IMMEDIATE 'DROP TABLE Material CASCADE CONSTRAINTS';  
EXCEPTION  
WHEN OTHERS THEN NULL;  
END;  
/
```

```
BEGIN
EXECUTE IMMEDIATE 'DROP TABLE Department CASCADE CONSTRAINTS';
EXCEPTION
WHEN OTHERS THEN NULL;
END;
/
```

```
BEGIN
EXECUTE IMMEDIATE 'DROP TABLE Supplier CASCADE CONSTRAINTS';
EXCEPTION
WHEN OTHERS THEN NULL;
END;
/
```

2. CREATE SUPPLIER TABLE

```
CREATE TABLE Supplier (
Supplier_ID NUMBER PRIMARY KEY,
Supplier_Name VARCHAR2(50),
Phone VARCHAR2(15),
Email VARCHAR2(50)
);
```

3. CREATE DEPARTMENT TABLE

```
CREATE TABLE Department (
Department_ID NUMBER PRIMARY KEY,
Department_Name VARCHAR2(50)
);
```

4. CREATE MATERIAL TABLE

```
CREATE TABLE Material (
Material_ID NUMBER PRIMARY KEY,
Material_Name VARCHAR2(50),
Category VARCHAR2(30),
Unit_Price NUMBER(10,2),
Stock_Quantity NUMBER,
Reorder_Level NUMBER,
Supplier_ID NUMBER REFERENCES Supplier(Supplier_ID)
);
```

5. CREATE MATERIAL REQUIREMENT TABLE

```
CREATE TABLE Material_Requirement (
Requirement_ID NUMBER PRIMARY KEY,
```

```
Department_ID NUMBER REFERENCES Department(Department_ID),  
Requirement_Date DATE,  
Requirement_Status VARCHAR2(20)  
);
```

6. CREATE REQUIREMENT DETAILS TABLE

```
CREATE TABLE Requirement_Details (  
Requirement_Detail_ID NUMBER PRIMARY KEY,  
Requirement_ID NUMBER REFERENCES Material_Requirement(Requirement_ID),  
Material_ID NUMBER REFERENCES Material(Material_ID),  
Required_Quantity NUMBER  
);
```

7. CREATE PURCHASE ORDER TABLE

```
CREATE TABLE Purchase_Order (  
PO_ID NUMBER PRIMARY KEY,  
Supplier_ID NUMBER REFERENCES Supplier(Supplier_ID),  
PO_Date DATE,  
PO_Status VARCHAR2(20)  
);
```

8. CREATE PURCHASE ORDER DETAILS TABLE

```
CREATE TABLE Purchase_Order_Details (  
PO_Detail_ID NUMBER PRIMARY KEY,  
PO_ID NUMBER REFERENCES Purchase_Order(PO_ID),  
Material_ID NUMBER REFERENCES Material(Material_ID),  
Ordered_Quantity NUMBER,  
Unit_Price NUMBER(10,2)  
);
```

9. INSERT SUPPLIER DATA

```
INSERT INTO Supplier VALUES  
(1,'ABC Suppliers','9876543210','abc@gmail.com');
```

```
INSERT INTO Supplier VALUES  
(2,'XYZ Traders','9876501234','xyz@gmail.com');
```

```
INSERT INTO Supplier VALUES  
(3,'Global Materials','9876512345','global@gmail.com');
```

10. INSERT DEPARTMENT DATA

INSERT INTO Department VALUES
(1,'Production');

INSERT INTO Department VALUES
(2,'Maintenance');

INSERT INTO Department VALUES
(3,'Electrical');

11. INSERT MATERIAL DATA

INSERT INTO Material VALUES
(101,'Steel Rod','Metal',250,80,20,1);

INSERT INTO Material VALUES
(102,'Copper Wire','Electrical',120,15,25,2);

INSERT INTO Material VALUES
(103,'Aluminium Sheet','Metal',180,60,15,3);

INSERT INTO Material VALUES
(104,'PVC Pipe','Plumbing',90,10,20,2);

INSERT INTO Material VALUES
(105,'Rubber Sheet','General',150,45,10,1);

12. INSERT MATERIAL REQUIREMENT DATA

INSERT INTO Material_Requirement VALUES
(1001,1,DATE '2026-09-20','Pending');

INSERT INTO Material_Requirement VALUES
(1002,2,DATE '2026-09-21','Approved');

INSERT INTO Material_Requirement VALUES
(1003,3,DATE '2026-09-22','Pending');

13. INSERT REQUIREMENT DETAILS

INSERT INTO Requirement_Details VALUES
(1,1001,101,30);

INSERT INTO Requirement_Details VALUES
(2,1001,102,40);

INSERT INTO Requirement_Details VALUES
(3,1002,102,20);

```
INSERT INTO Requirement_Details VALUES  
(4,1002,103,15);
```

```
INSERT INTO Requirement_Details VALUES  
(5,1003,104,25);
```

```
INSERT INTO Requirement_Details VALUES  
(6,1003,101,10);
```

14. INSERT PURCHASE ORDER DATA

```
INSERT INTO Purchase_Order VALUES  
(5001,1,DATE '2026-09-23','Pending');
```

```
INSERT INTO Purchase_Order VALUES  
(5002,2,DATE '2026-09-24','Approved');
```

```
INSERT INTO Purchase_Order VALUES  
(5003,3,DATE '2026-09-25','Pending');
```

15. INSERT PURCHASE ORDER DETAILS

```
INSERT INTO Purchase_Order_Details VALUES  
(1,5001,101,20,250);
```

```
INSERT INTO Purchase_Order_Details VALUES  
(2,5001,105,15,150);
```

```
INSERT INTO Purchase_Order_Details VALUES  
(3,5002,102,50,120);
```

```
INSERT INTO Purchase_Order_Details VALUES  
(4,5002,104,30,90);
```

```
INSERT INTO Purchase_Order_Details VALUES  
(5,5003,103,20,180);
```

```
COMMIT;
```

16. DISPLAY ALL MATERIALS

```
SELECT *  
FROM Material;
```

OUTPUT:

MATERIAL_ID MATERIAL_NAME CATEGORY UNIT_PRICE

101 Steel Rod Metal 250
102 Copper Wire Electrical 120
103 Aluminium Sheet Metal 180
104 PVC Pipe Plumbing 90
105 Rubber Sheet General 150

STOCK_QUANTITY REORDER_LEVEL SUPPLIER_ID

80 20 1
15 25 2
60 15 3
10 20 2
45 10 1

17. MATERIALS BELOW REORDER LEVEL

```
SELECT
Material_ID,
Material_Name,
Stock_Quantity,
Reorder_Level
FROM Material
WHERE Stock_Quantity < Reorder_Level;
```

OUTPUT:

MATERIAL_ID MATERIAL_NAME STOCK_QUANTITY REORDER_LEVEL

102 Copper Wire 15 25
104 PVC Pipe 10 20

18. MATERIAL AND SUPPLIER DETAILS

```
SELECT
m.Material_ID,
```

```
m.Material_Name,  
m.Category,  
s.Supplier_Name  
FROM Material m  
JOIN Supplier s  
ON m.Supplier_ID = s.Supplier_ID;
```

OUTPUT:

MATERIAL_ID MATERIAL_NAME CATEGORY SUPPLIER_NAME

101 Steel Rod Metal ABC Suppliers
102 Copper Wire Electrical XYZ Traders
103 Aluminium Sheet Metal Global Materials
104 PVC Pipe Plumbing XYZ Traders
105 Rubber Sheet General ABC Suppliers

19. TOTAL STOCK VALUE

```
SELECT  
SUM(Unit_Price * Stock_Quantity) AS Total_Stock_Value  
FROM Material;
```

OUTPUT:

TOTAL_STOCK_VALUE

40250

20. MATERIAL COUNT BY CATEGORY

```
SELECT  
Category,  
COUNT(*) AS Number_Of_Materials  
FROM Material  
GROUP BY Category;
```

OUTPUT:

CATEGORY NUMBER_OF_MATERIALS

Metal 2
Electrical 1
Plumbing 1
General 1

21. REQUIREMENT DETAILS

```
SELECT
mr.Requirement_ID,
d.Department_Name,
mr.Requirement_Date,
mr.Requirement_Status
FROM Material_Requirement mr
JOIN Department d
ON mr.Department_ID = d.Department_ID;
```

OUTPUT:

REQUIREMENT_ID	DEPARTMENT_NAME	REQUIREMENT_DATE	STATUS
----------------	-----------------	------------------	--------

1001	Production	20-SEP-26	Pending
1002	Maintenance	21-SEP-26	Approved
1003	Electrical	22-SEP-26	Pending

22. REQUIRED MATERIAL QUANTITIES

```
SELECT
m.Material_Name,
SUM(rd.Required_Quantity) AS Total_Required_Quantity
FROM Requirement_Details rd
JOIN Material m
ON rd.Material_ID = m.Material_ID
GROUP BY m.Material_Name;
```

OUTPUT:

MATERIAL_NAME TOTAL_REQUIRED_QUANTITY

Steel Rod 40
Copper Wire 60
Aluminium Sheet 15
PVC Pipe 25

23. MOST EXPENSIVE MATERIAL

```
SELECT
Material_ID,
Material_Name,
Unit_Price
FROM Material
WHERE Unit_Price = (
SELECT MAX(Unit_Price)
FROM Material
);
```

OUTPUT:

MATERIAL_ID MATERIAL_NAME UNIT_PRICE

101 Steel Rod 250

24. PURCHASE ORDER DETAILS

```
SELECT
po.PO_ID,
s.Supplier_Name,
po.PO_Date,
po.PO_Status
FROM Purchase_Order po
JOIN Supplier s
ON po.Supplier_ID = s.Supplier_ID;
```

OUTPUT:

PO_ID SUPPLIER_NAME PO_DATE PO_STATUS

5001 ABC Suppliers 23-SEP-26 Pending
5002 XYZ Traders 24-SEP-26 Approved
5003 Global Materials 25-SEP-26 Pending

25. PURCHASE ORDER MATERIAL DETAILS

```
SELECT
po.PO_ID,
s.Supplier_Name,
m.Material_Name,
pod.Ordered_Quantity,
pod.Unit_Price
FROM Purchase_Order_Details pod
JOIN Purchase_Order po
ON pod.PO_ID = po.PO_ID
JOIN Supplier s
ON po.Supplier_ID = s.Supplier_ID
JOIN Material m
ON pod.Material_ID = m.Material_ID;
```

OUTPUT:

PO_ID SUPPLIER_NAME MATERIAL_NAME ORDERED_QTY UNIT_PRICE

5001 ABC Suppliers Steel Rod 20 250
5001 ABC Suppliers Rubber Sheet 15 150
5002 XYZ Traders Copper Wire 50 120
5002 XYZ Traders PVC Pipe 30 90
5003 Global Materials Aluminium Sheet 20 180

26. UPDATE STOCK QUANTITY

```
UPDATE Material
SET Stock_Quantity = Stock_Quantity + 20
WHERE Material_ID = 102;
```

COMMIT;

27. DISPLAY UPDATED MATERIAL

```
SELECT *  
FROM Material  
WHERE Material_ID = 102;
```

OUTPUT:

MATERIAL_ID	MATERIAL_NAME	CATEGORY	UNIT PRICE
-------------	---------------	----------	------------

102	Copper Wire	Electrical	120
-----	-------------	------------	-----

STOCK_QUANTITY	REORDER_LEVEL	SUPPLIER_ID
----------------	---------------	-------------

35	25	2
----	----	---

28. DELETE A PENDING PURCHASE ORDER

```
DELETE FROM Purchase_Order_Details  
WHERE PO_ID = 5003;
```

```
DELETE FROM Purchase_Order  
WHERE PO_ID = 5003;
```

```
COMMIT;
```

29. DISPLAY REMAINING PURCHASE ORDERS

```
SELECT *  
FROM Purchase_Order;
```

OUTPUT:

PO_ID	SUPPLIER_ID	PO_DATE	PO_STATUS
-------	-------------	---------	-----------

5001	1	23-SEP-26	Pending
------	---	-----------	---------

5002	2	24-SEP-26	Approved
------	---	-----------	----------

Purchase Order 5003 has been deleted.

30. VERIFY PURCHASE ORDER DETAILS

```
SELECT *  
FROM Purchase_Order_Details;
```

OUTPUT:

PO_DETAIL_ID	PO_ID	MATERIAL_ID	ORDERED_QUANTITY	UNIT_PRICE
--------------	-------	-------------	------------------	------------

1	5001	101	20	250
---	------	-----	----	-----

2	5001	105	15	150
---	------	-----	----	-----

3	5002	102	50	120
---	------	-----	----	-----

4	5002	104	30	90
---	------	-----	----	----

END OF INVENTORY CONTROL SYSTEM